

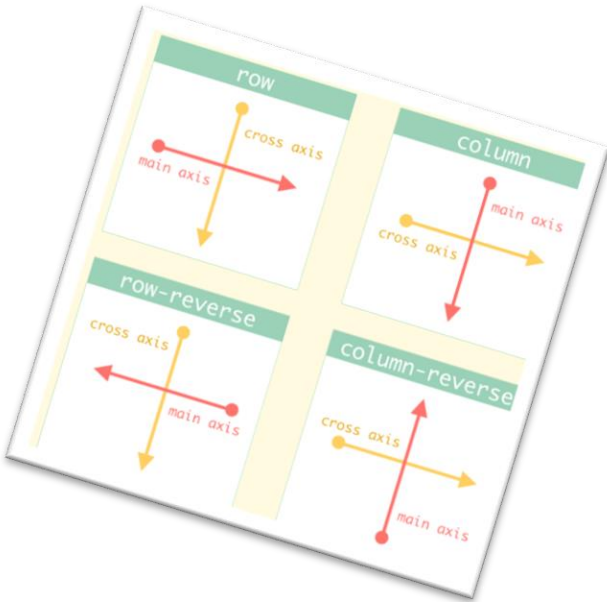
Preparation before next session



Next session will include a quiz !

PART 1 – READ / WATCH

1. [Read our amazing book !](#)
2. [Watch video Flex in 10 minutes !](#)
3. [Check the Flexbox handout !](#)



PART 2 – GAME

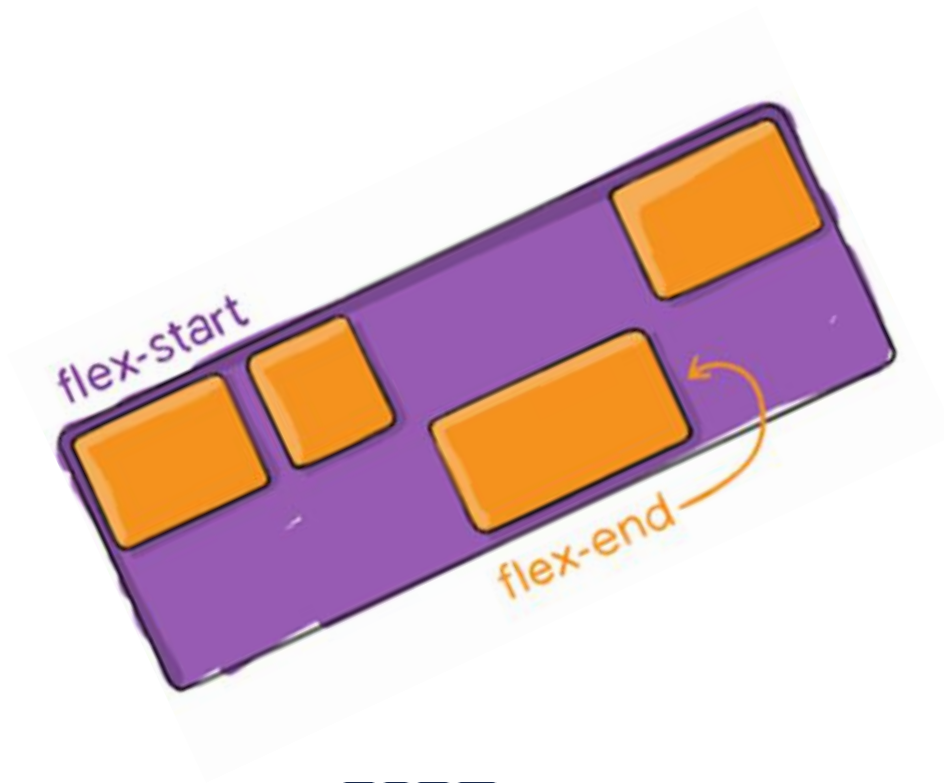
Go as far as you can on this game

<https://flexboxfroggy.com>



WEB DESIGN

W4-S1 – Flexbox





Part 1 – Objectives

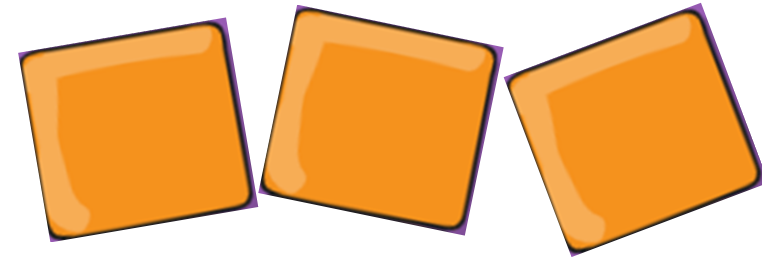


FLEX CONTAINER



- ✓ flex-direction
- ✓ justify-content
- ✓ align-items
- flex-wrap

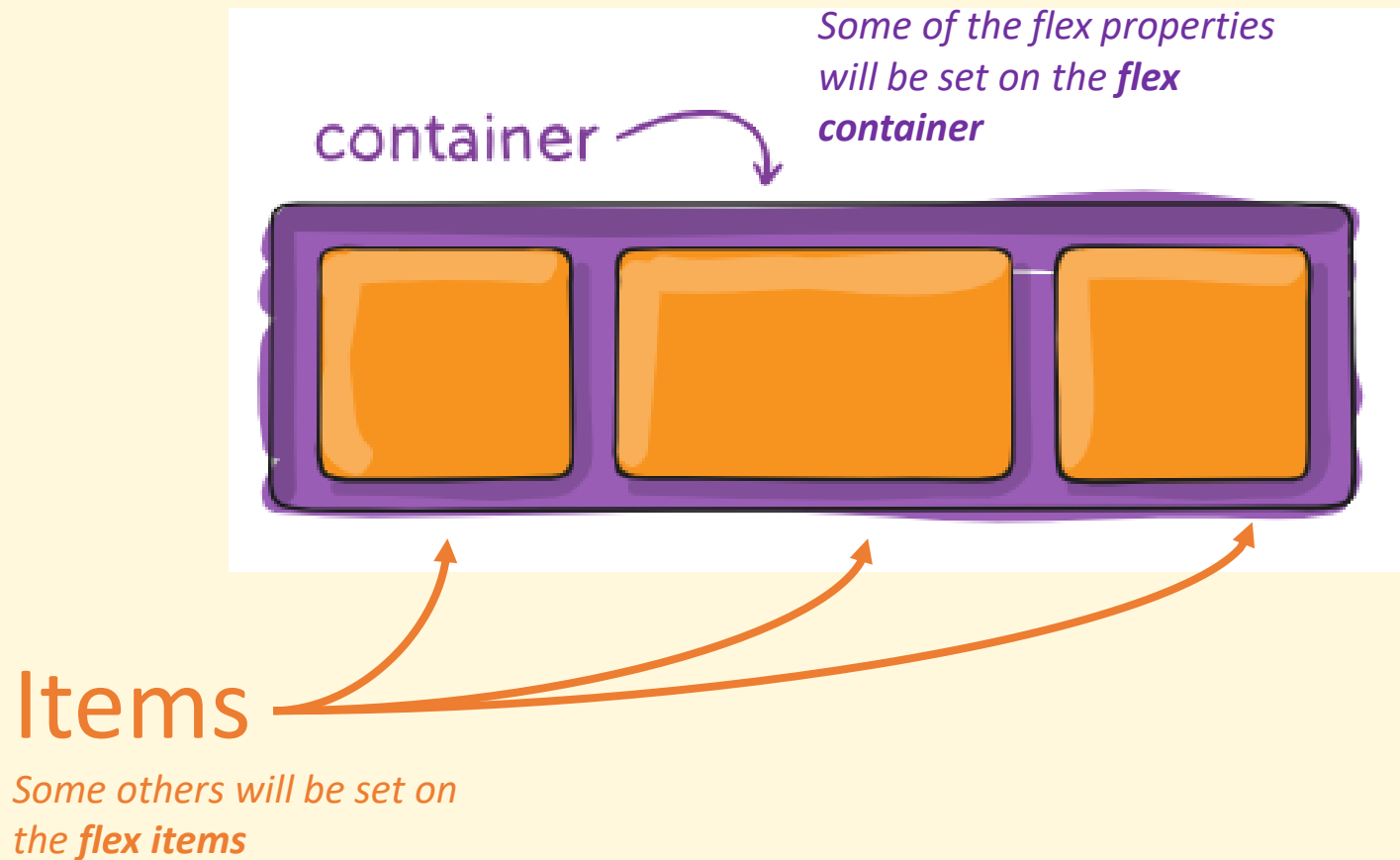
FLEX ITEMS



- ✓ align-self
- flex-grow, flex-shrink flex-basis
- auto margins in flex items

Flexbox terminology

Make a difference between the flex **containers** and the flex **items**

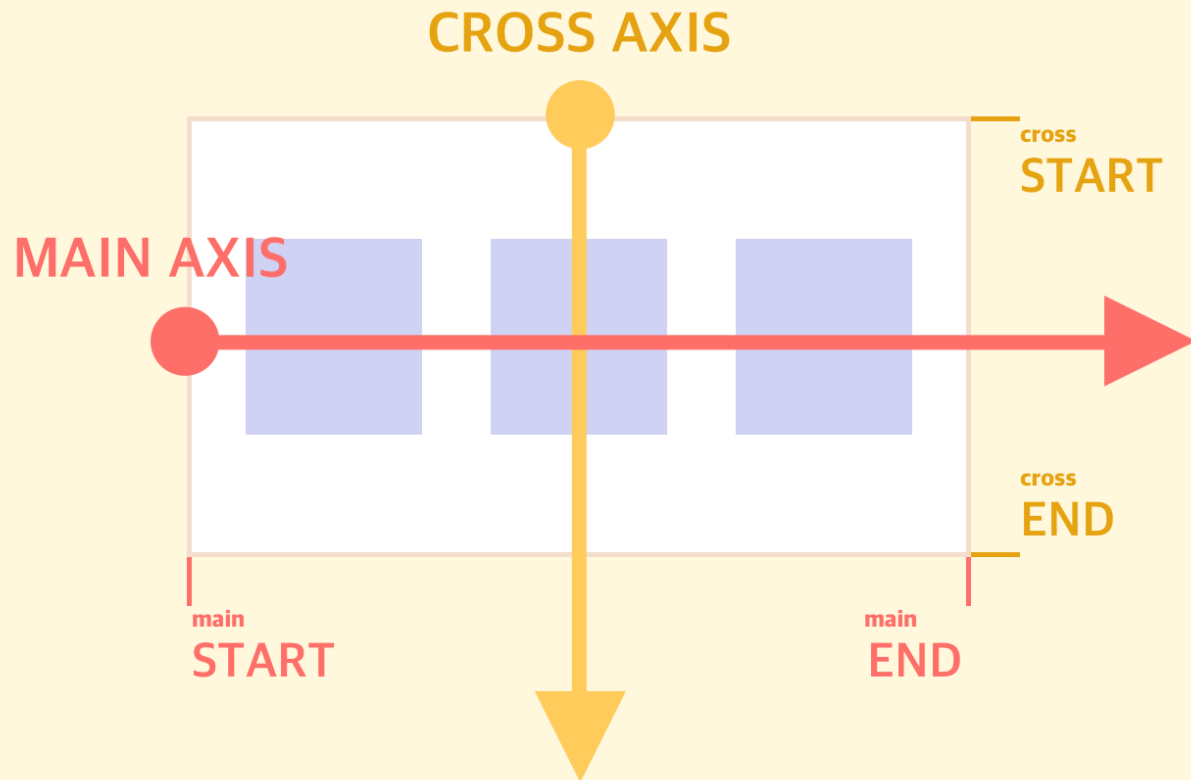


```
<div class="container">  
  <div>Item 1</div>  
  <div>Item 2</div>  
  <div>Item 3</div>  
</div>
```

```
.container {  
  display: flex;  
}
```

Flexbox terminology

Flexbox operates in a 2 axes systems: the **main** and the **cross** axis

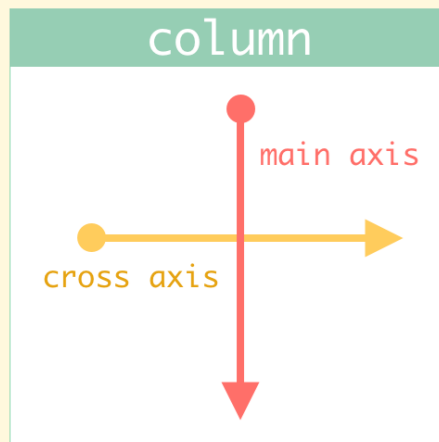
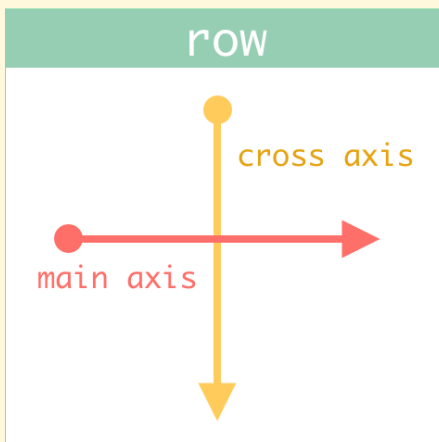


- The main axis is your defining direction of **how your flex items are placed in the flex container**
- The cross axis is in the direction that's **perpendicular** to the main axis.

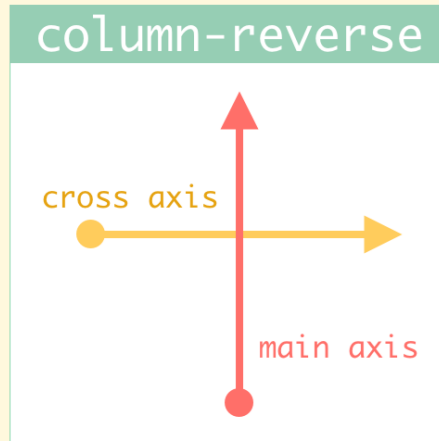
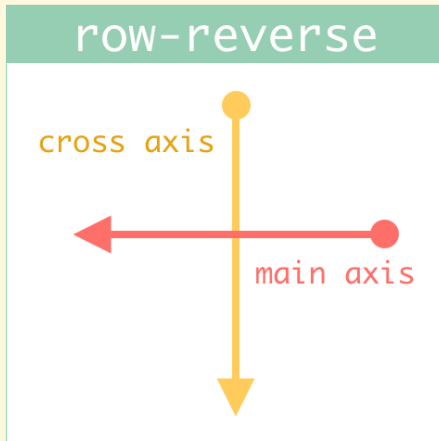
Like X, Y axis in Math !

Flex Direction

You can **change the direction** of the main axis with flex-direction



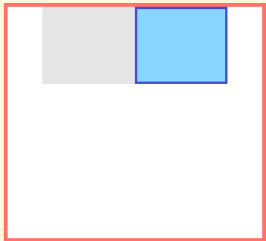
```
.container {  
  display: flex;  
  flex-direction : column;  
}
```



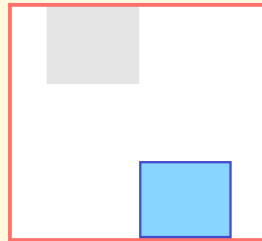
Align self

Break the alignment rule !!!

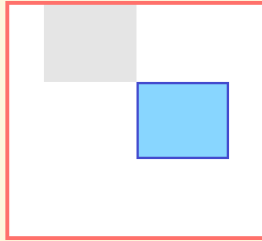
flex-start



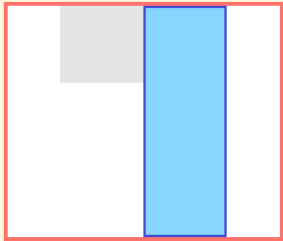
flex-end



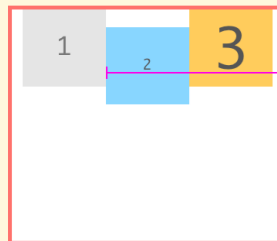
center



stretch



baseline



- The thing with align-items is that it forces ALL of the flex items to play with the rules.
- If you want one of them to break the rule, you can use align-self.

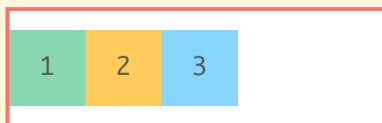
```
#myItem {  
  align-self: flex-start  
}
```

Alignment Properties

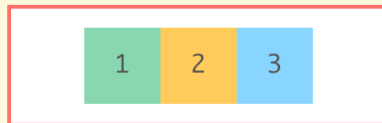
Align items long the **main axis**

```
.container {  
  display: flex;  
  justify-items: flex-end;  
}
```

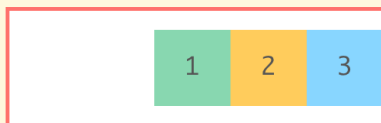
flex-start



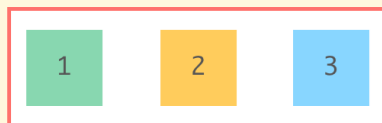
center



flex-end



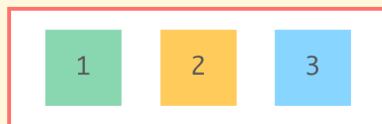
space-around



space-between



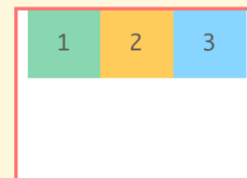
space-evenly



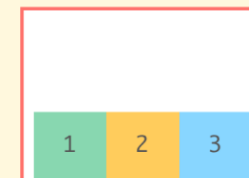
Align items long the **cross axis**

```
.container {  
  display: flex;  
  align-items: flex-end;  
}
```

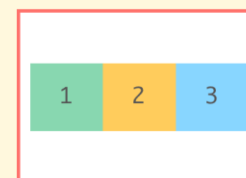
flex-start



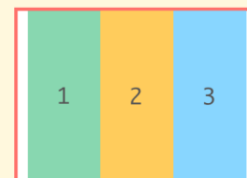
flex-end



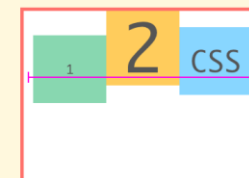
center



stretch



baseline



Alignment

What is the difference between **space-around** and **space-evenly**?

space-around



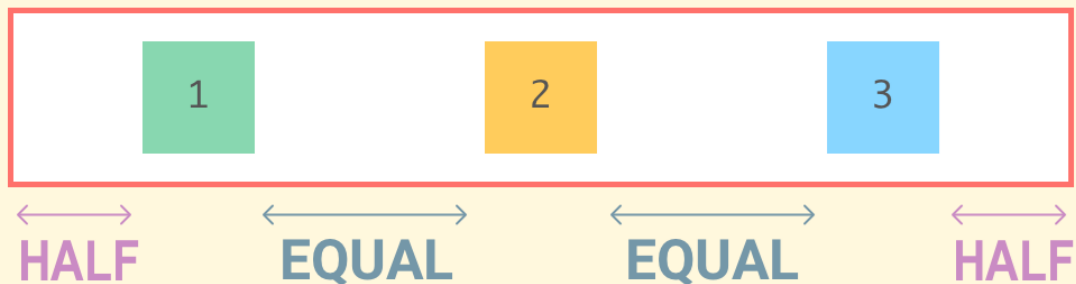
space-evenly



Alignment

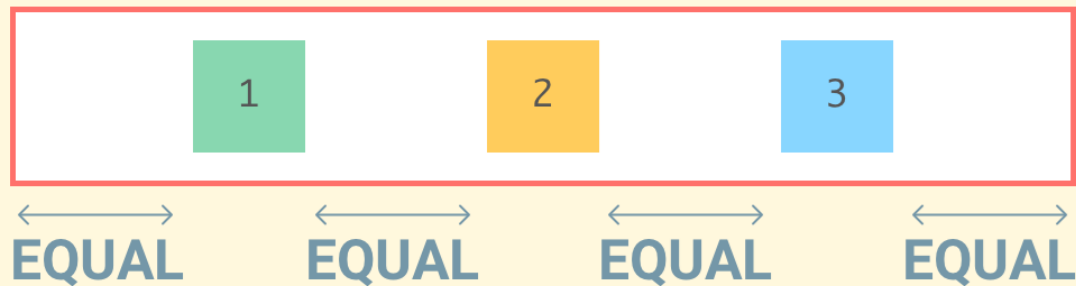
What is the difference between **space-around** and **space-evenly**?

space-around



In space-around, only the inner items will have equal spacing in between each other. The first and last item will only be allocated half the spacing.

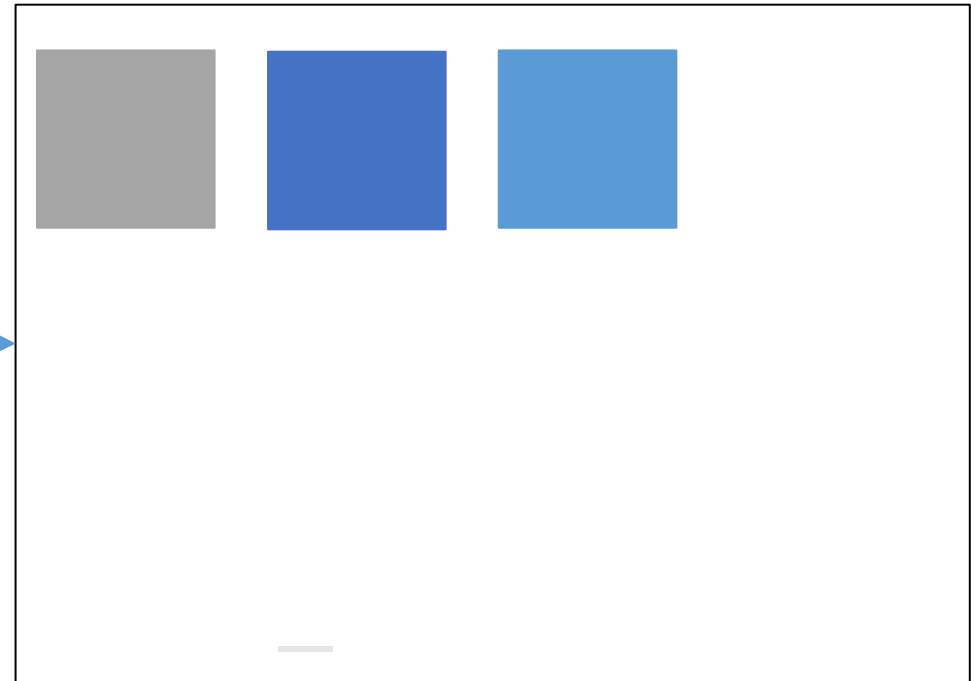
space-evenly



In space-evenly, the empty space in between the flex items is always equal.

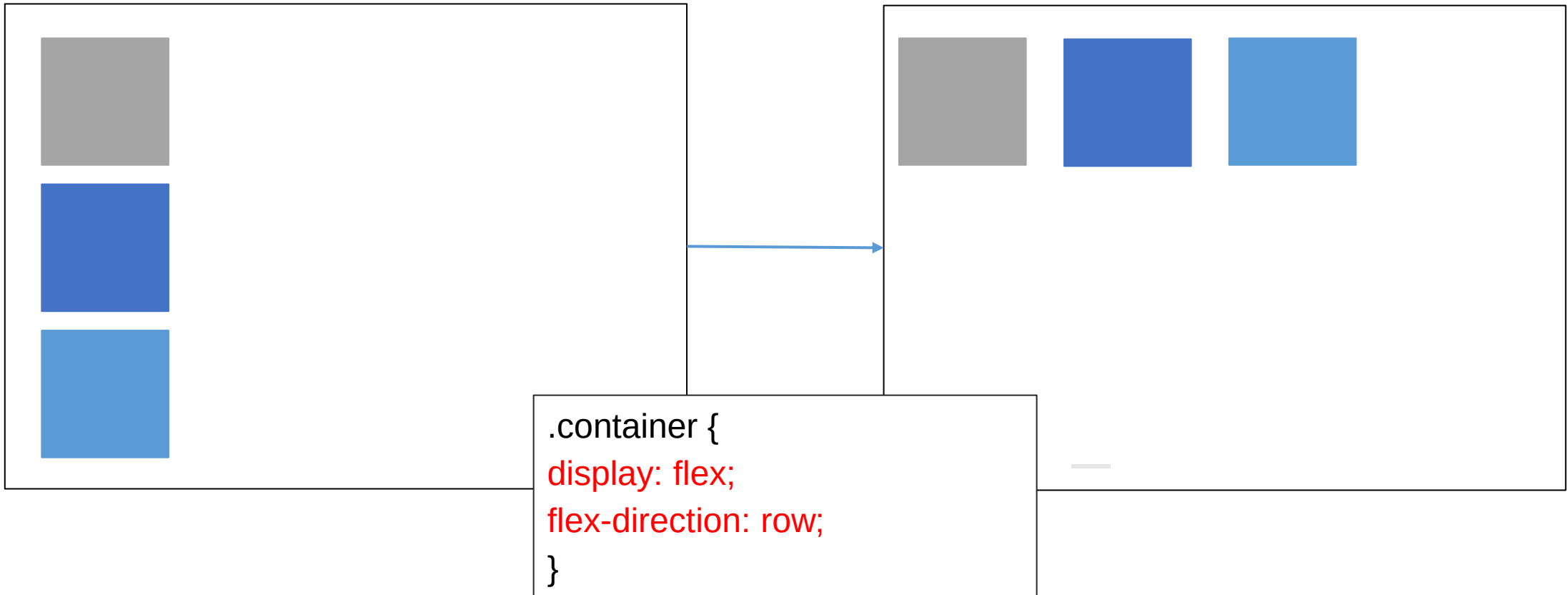
What properties are applied here?

```
<div class="container">
```



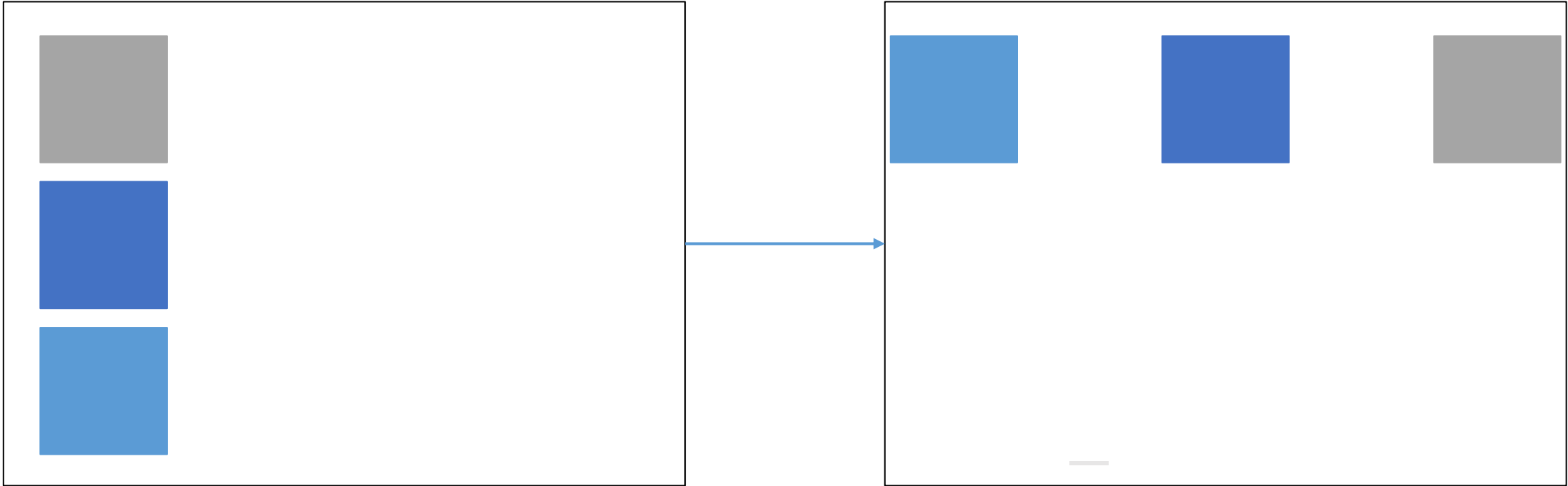
What properties are applied here?

```
<div class="container">
```



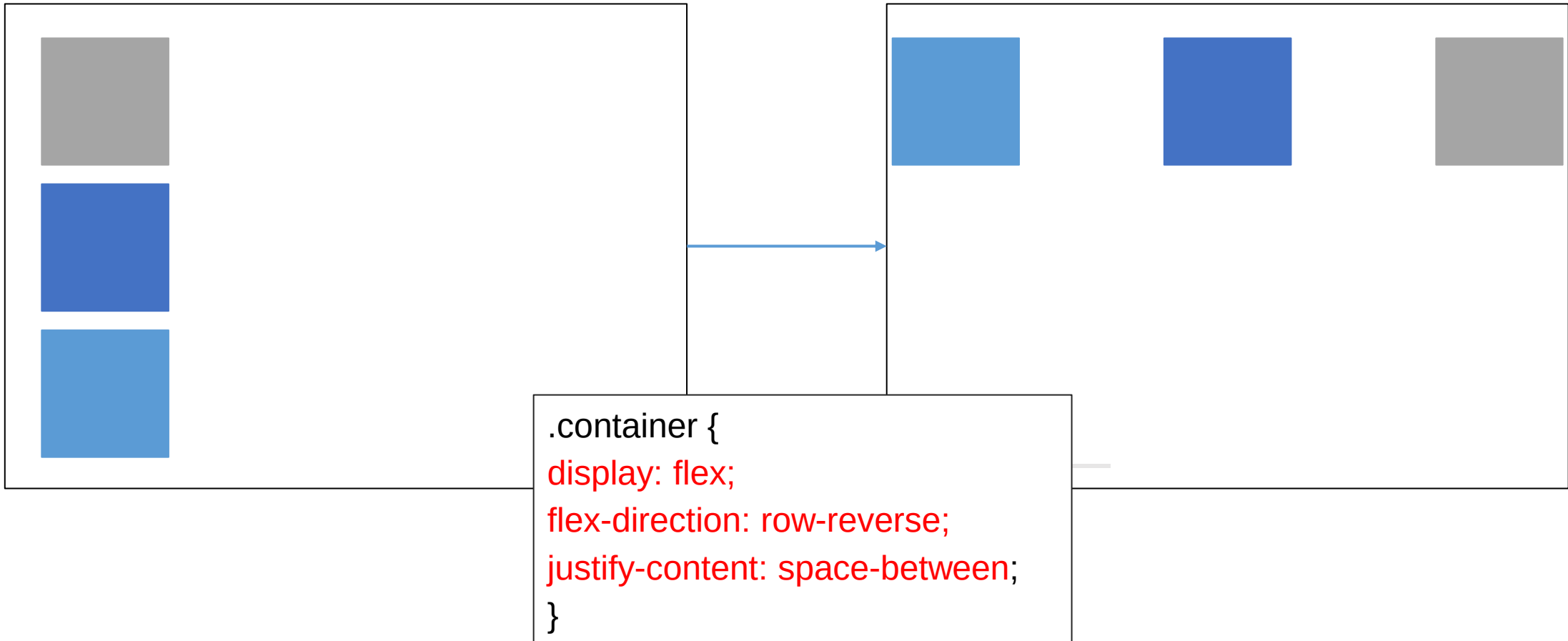
What properties are applied here?

```
<div class="container">
```



What properties are applied here?

```
<div class="container">
```





20 MIN

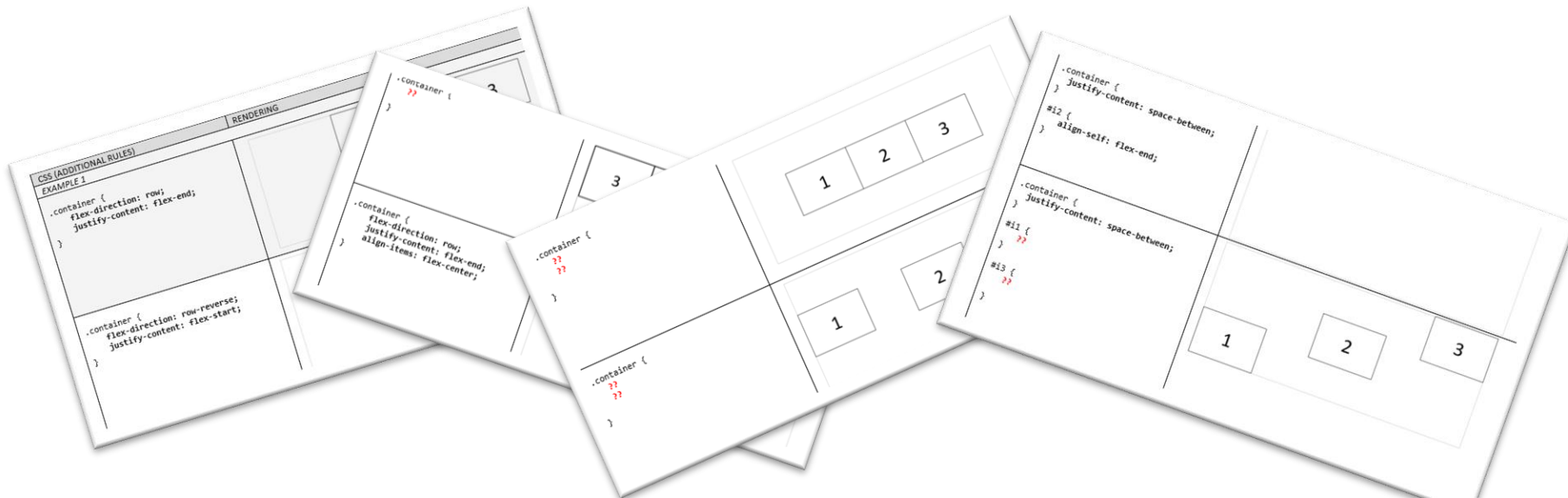
Activity 1

We want you to **understand** the following properties and their possible values:

PROPERTY	POSSIBLE VALUES
flex-direction	row, column, row-reverse, column-reverse
justify-content	flex-start, flex-end, center, space-between, space-around
align-items	flex-start, flex-end, center, baseline
align-self	flex-start, flex-end, center, baseline

READ THE [GUIDE](#) !

- Complete first individually the 10 challenges on paper
- Then share by group of 3 students





Part 2 – Objectives



FLEX CONTAINER



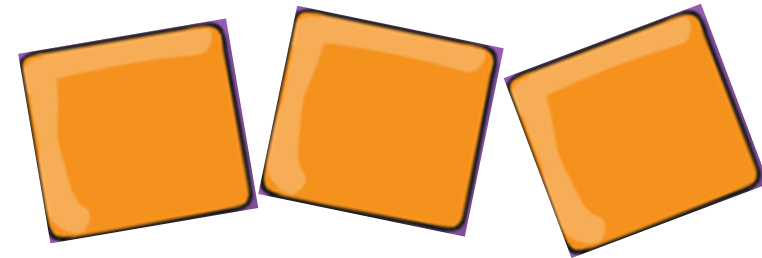
flex-direction

justify-content

align-items

✓ flex-wrap

FLEX ITEMS



align-self

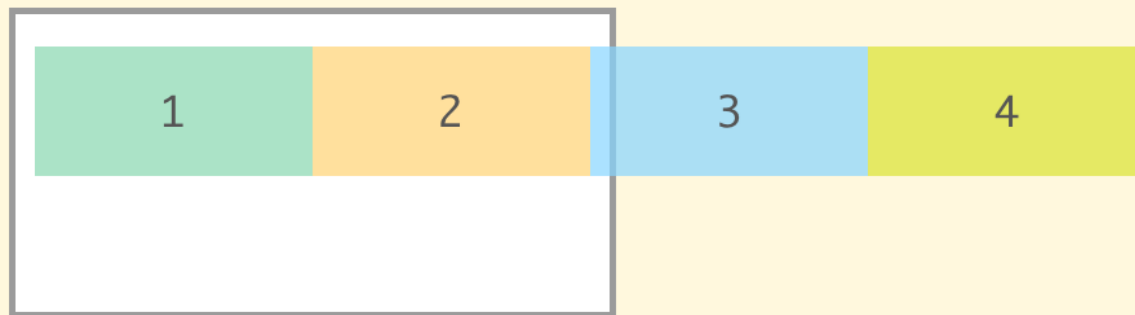
✓ flex-grow, flex-shrink flex-basis

✓ auto margins in flex items

Flex Wrap

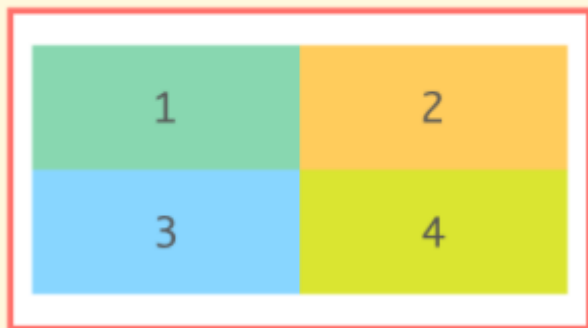
- By default, flex items will try to shrink itself to fit onto one line, in other words, **no wrap**.

original size



- To spread items on multiple lines in the containers, you can turn on wrap or wrap-reverse

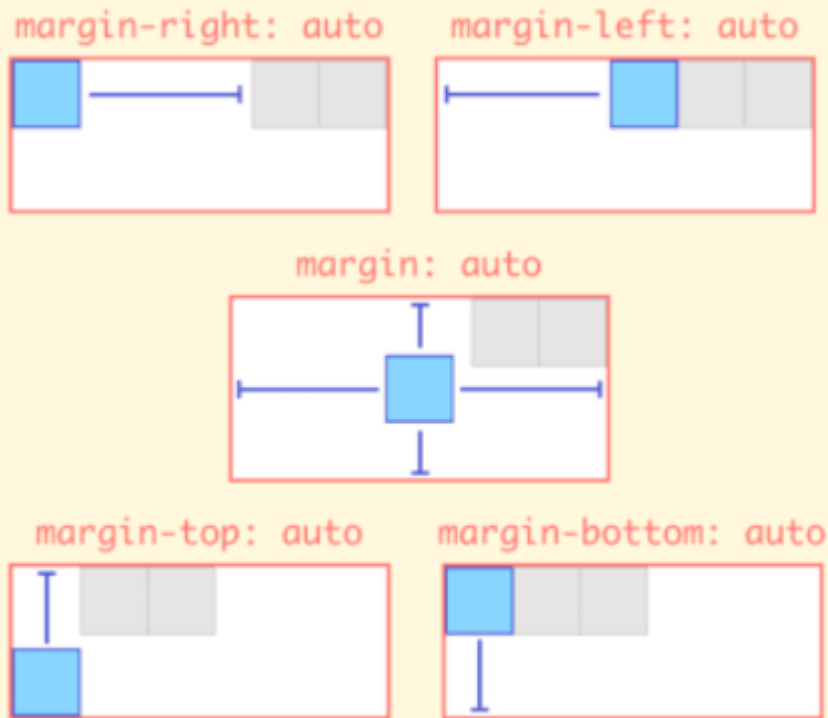
wrap



```
.container {  
  display: flex;  
  flex-wrap: wrap  
}
```

Auto Margin in Flex

Bonus content! Another way to align Flexbox child elements is to use auto margins



Although this isn't a Flexbox property, it's still important to be aware of it because it has a very [interesting relationship with Flexbox](#)

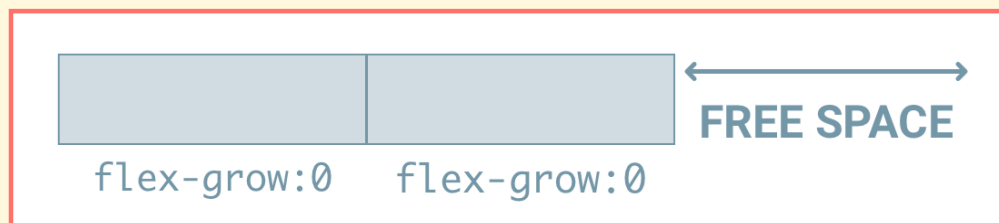
```
#item2 {  
  margin-left : auto;  
}
```

ITEMS

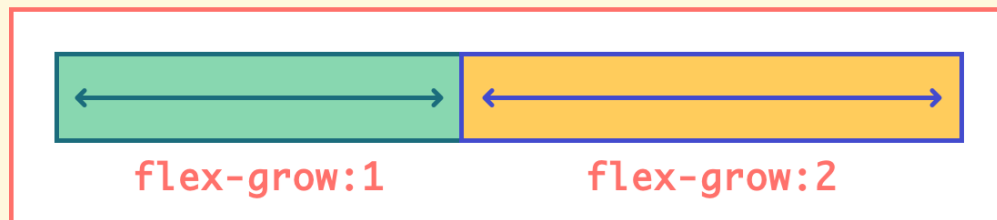
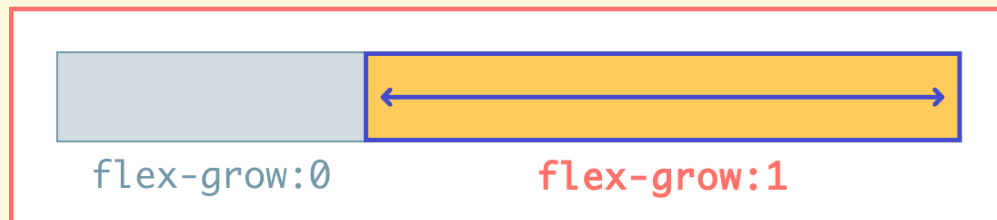
Flex Grow & Shrink

Flexbox is great for responsive design!

- By default growing or shrinking is not enabled:



- The `flex-grow` property allows our flex item to grow if necessary



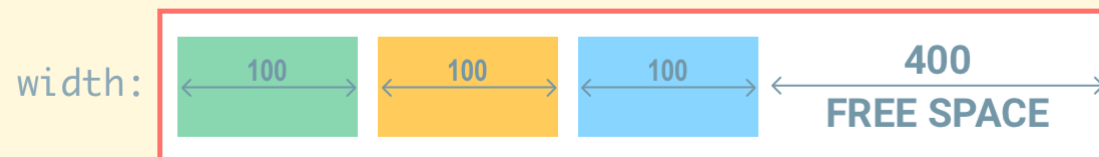
```
#item2 {  
  flex-grow : 1;  
}
```

ITEMS

Flex Grow & Shrink

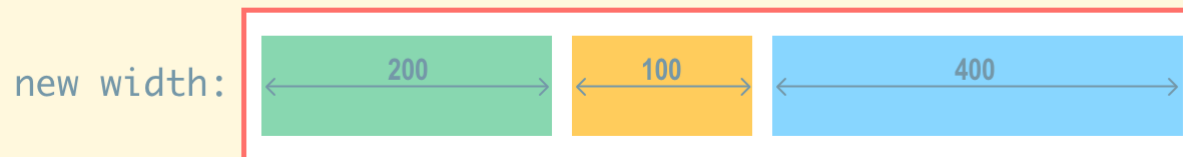
Let's see the calculation in details

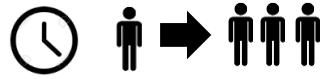
$$\text{new width} = \left(\frac{\text{flex grow}}{\text{total flex grow}} \times \text{free space} \right) + \text{width}$$



calculation:

$$\left(\frac{\textcircled{1}}{4} \times 400 \right) + 100 \quad \left(\frac{\textcircled{0}}{4} \times 400 \right) + 100 \quad \left(\frac{\textcircled{3}}{4} \times 400 \right) + 100$$





20 MIN

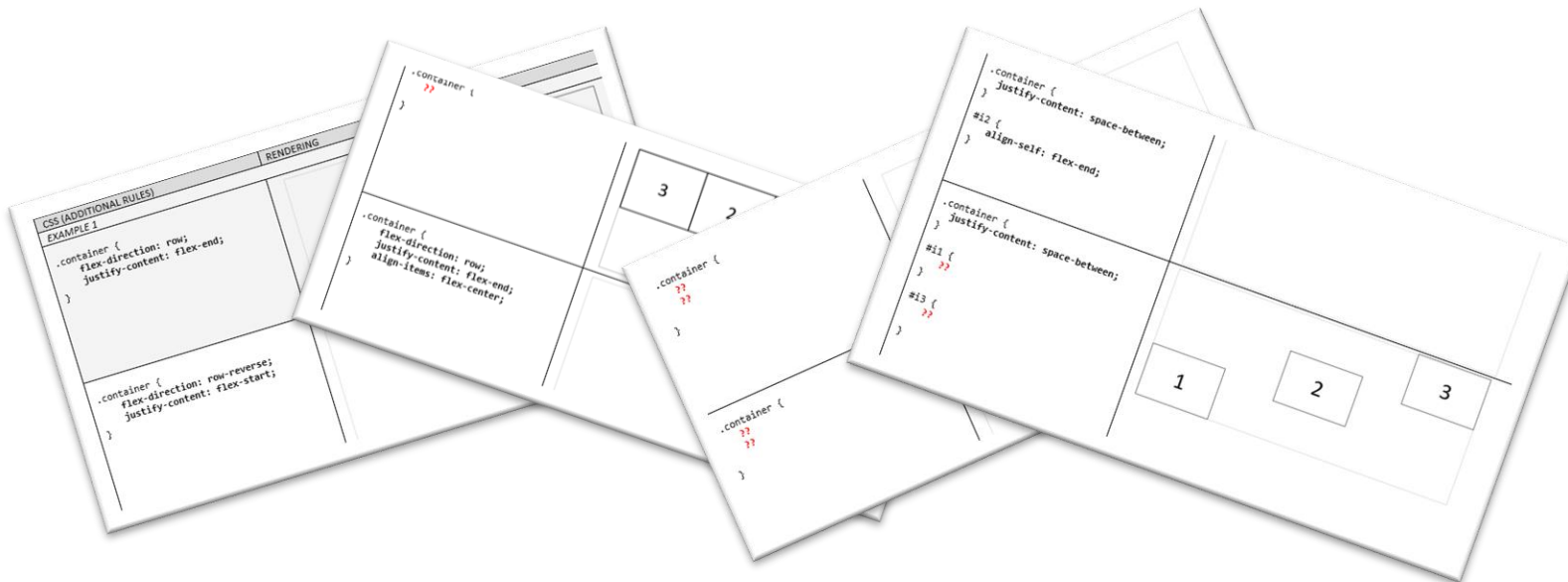
Activities 2 and 3

We want you to **understand** the following properties and their possible values:

PROPERTIES	POSSIBLE VALUES
flex-wrap	nowrap, wrap, wrap-reverse
flex-grow	An integer ≥ 0
flex-shrink	An integer ≥ 0
flex-basis	An integer ≥ 0

READ THE [GUIDE](#) !

- Complete first individually the activities 2 and 3 on paper
- Then share by group of 3 students





Congratulations !



You should now master those concepts

DIRECTION

flex-direction

flex-wrap

flex-flow

order

ALIGNMENT

justify-content

align-items

align-content

align-self

SIZE

flex-grow

flex-shrink

flex-basis

flex

● PARENT

● CHILD

3-2-1 Challenge

- ✓ List three things you **learned** today.
- ✓ List two **questions** you still have.
- ✓ List one aspect of the lesson or topic you **enjoyed**.



After this session

1 Review the theory

[Read again the amazing book](#)
[Check the Flexbox handout](#)

2 Try the Flex Adventure !

[The Flex Adventure](#)

3 Write your OWN code

To master the concepts

